LHCb Open Data Analysis

A CERN Open Data Analysis of Three-Body B-Meson Decays

Invariant-Mass Reconstruction, Signal Classification, CP-Asymmetry Diagnostics, and Dalitz-Plot Studies

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Portfolio report for experimental particle-physics PhD applications

Abstract

This report presents a compact but realistic high-energy-physics analysis workflow based on LHCb CERN Open Data. Three-body B-meson candidates are reconstructed from charged-hadron tracks under a kaon mass hypothesis. The workflow uses ROOT input files, Python-based columnar data processing, invariant-mass reconstruction, event selection, a sideband-labelled Random Forest classifier, a raw charge-asymmetry diagnostic, and Dalitz-plot visualisations. The report emphasizes reproducibility, modular scientific code, and analysis reasoning suitable for an early-stage experimental particle-physics project.

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1 Introduction

The Standard Model of particle physics describes the known elementary particles and their interactions with remarkable precision. Nevertheless, several observed features of the Universe remain only partially understood. One of the most important is the matter–antimatter asymmetry: cosmological observations indicate that the visible Universe is dominated by matter, while the known laws of particle physics contain only limited sources of charge-parity (CP) violation. Precision studies of flavour-changing processes are therefore a central part of modern particle physics.

Flavour physics studies transitions between different quark flavours. In the heavy-flavour sector, hadrons containing beauty quarks provide especially rich laboratories. The large mass of the beauty quark allows many decay modes, including multibody final states in which resonant and non-resonant amplitudes can interfere. Such interference can generate local CP-asymmetric effects in restricted regions of phase space, even when the integrated asymmetry is small.

B mesons are bound states containing a beauty quark or antiquark and a lighter valence quark. Charged B mesons, denoted B^+ and B^-, form a natural pair of charge-conjugate systems. By comparing the observed yields or distributions of B^+ and B^- candidates, one can construct raw asymmetry observables. In a simplified counting analysis, the integrated asymmetry is written as

$$A_{\mathrm{CP}}^{\mathrm{raw}} = \frac{N(B^-) - N(B^+)}{N(B^-) + N(B^+)}, \tag{1}$$

where $N(B^+)$ and $N(B^-)$ are observed candidate counts. A full physics measurement must correct this raw quantity for background contributions, detector-induced charge asymmetries, production asymmetries, trigger effects, and selection efficiencies. The present project treats Eq. 1 as an analysis diagnostic rather than as a final CP-violation measurement.

This report documents a reproducible analysis workflow built with CERN Open Data [1]. The project focuses on a sample of reconstructed LHCb three-hadron B candidates stored in ROOT files. The analysis reconstructs the B-candidate invariant mass, applies simple physics-motivated selections, fits the mass peak, trains a signal/background classifier using sideband labels, separates charge-conjugate candidates, and produces Dalitz-plot distributions. The goal is to demonstrate a modern high-energy-physics (HEP) workflow using transparent scientific reasoning and modular Python code.

The analysis is intentionally compact. It is not presented as a publishable LHCb measurement, but as a polished student-level project that mirrors the structure of a real experimental analysis: data access, reconstruction, selection, modelling, validation plots, and interpretation of limitations. This makes it suitable as a technical portfolio project for applications in experimental particle physics.

2 The LHCb Experiment

The Large Hadron Collider beauty experiment (LHCb) is a forward spectrometer at the CERN Large Hadron Collider designed for precision studies of hadrons containing beauty and charm quarks [2]. Proton–proton collisions at the LHC produce heavy quarks predominantly in the forward and backward regions. LHCb exploits this production topology by covering a forward angular region, providing excellent tracking, vertexing, momentum resolution, and particle identification.

The detector design is particularly well suited to heavy-flavour analyses. B hadrons have lifetimes long enough to travel measurable distances before decaying. This displaced-decay topology is identified by reconstructing primary vertices from the proton–proton interaction and secondary vertices from the B-hadron decay products. The distance between these vertices and the quality of the reconstructed secondary vertex are powerful handles for suppressing prompt combinatorial background.

Charged-particle tracking is performed with a system of silicon and tracking stations around a dipole magnet. The curvature of charged-particle trajectories in the magnetic field provides a momentum measurement and determines the charge sign. In the present analysis, the track momentum components are used directly to build four-vectors for the three daughter hadrons. The charge information is used later to classify candidates as B^+ or B^-.

Particle identification (PID) is another central ingredient. LHCb uses ring-imaging Cherenkov detectors to distinguish kaons, pions, and protons over a broad momentum range. The open-data ntuples used in this project contain per-track PID-related variables such as kaon and pion probability-like observables. A simple kaon-PID selection is applied to build a clean demonstration sample under a $B \to KKK$ mass hypothesis.

In a complete LHCb analysis, detector effects are modelled with simulation, calibration samples, and data-driven corrections. This project keeps the detector treatment deliberately simple, but it preserves the main conceptual ingredients: displaced B vertices, charged-track reconstruction, particle identification, and magnet-polarity awareness.

3 Dataset and Software Framework

3.1 ROOT input files

The project uses CERN Open Data ROOT files containing reconstructed B-candidate ntuples. The main file used in the notebooks is

`data/raw/B2HHH_MagnetDown.root`.

Additional files for future extensions are stored alongside it:

`B2HHH_MagnetUp.root`, `PhaseSpaceSimulation.root`.

The presence of both magnet polarities is important for realistic charge-asymmetry studies, because reversing the magnetic field helps control detector-induced left–right and charge-dependent reconstruction effects. The present report focuses on the MagnetDown sample for a compact demonstration.

The data are stored in the ROOT format [3], a standard file and object format in high-energy physics. The relevant event information is contained in a tree named `DecayTree`. A ROOT `TTree` stores columnar data: each branch corresponds to one observable, such as a momentum component, charge, PID response, vertex quality, or flight distance. This format is efficient for large event samples because only the branches needed for a given analysis step need to be read.

3.2 Python analysis workflow

The analysis is implemented as a Python workflow, with reusable functions under `src/` and exploratory or result-producing notebooks under `notebooks/`. The main software components are listed in Table 1.

Table 1: Main software tools used in the analysis workflow.

Tool	Role in the analysis
ROOT	Standard HEP file format and ntuple storage
`uproot`	Reading ROOT TTrees directly in Python
`pandas`	Tabular event manipulation and selections
`numpy`	Vectorised numerical calculations
`matplotlib`	Publication-style figures
`scikit-learn`	Random Forest classifier and performance metrics

The branch set used throughout the analysis includes the momentum components `H1_PX`, `H1_PY`, `H1_PZ` through `H3_PZ`, the three track charges, PID variables such as `H1_ProbK`, and B-candidate quality variables such as `B_VertexChi2` and `B_FlightDistance`. A shared selection helper loads these branches, reconstructs the B-candidate mass, and applies the common baseline cuts. This avoids duplicated logic between the mass, machine-learning, CP-asymmetry, and Dalitz notebooks.

The workflow is intentionally modular. The main reusable components are the invariant-mass reconstruction module `src/reconstruction/invariant_mass.py`, the mass-plotting helpers in `src/plotting/mass_plots.py`, the signal/background utilities in `src/ml/signal_background.py`, and the Dalitz-kinematics module `src/physics/dalitz.py`. This separation mirrors the structure of a maintainable HEP analysis, where physics definitions, plotting conventions, and machine-learning utilities should be reusable across notebooks.

4 Invariant Mass Reconstruction

4.1 Relativistic mass formula

The key reconstructed observable is the invariant mass of the three-hadron B candidate. For a system of particles with total energy E and total momentum $\vec{p}$, the invariant mass is defined by

$$m^2c^4 = E^2 - |\vec{p}|^2c^2. \tag{2}$$

In high-energy-physics units with $c = 1$, this becomes

$$m^2 = E^2 - p_x^2 - p_y^2 - p_z^2. \tag{3}$$

For each daughter track, the analysis uses a charged-kaon mass hypothesis,

$$m_K = 493.677\,\mathrm{MeV}/c^2, \tag{4}$$

and reconstructs the energy as

$$E_i = \sqrt{p_{x,i}^2 + p_{y,i}^2 + p_{z,i}^2 + m_K^2}. \tag{5}$$

The B-candidate four-vector is then the sum of the three daughter four-vectors:

$$E_B = \sum_{i=1}^{3} E_i, \qquad \vec{p}_B = \sum_{i=1}^{3} \vec{p}_i. \tag{6}$$

Substituting these quantities into Eq. 3 gives the reconstructed B-candidate mass. The calculation is performed event-by-event with vectorised `numpy` operations.

4.2 Mass spectrum before and after selection

Figure 1 shows the reconstructed three-kaon mass spectrum before the baseline event selection. The distribution contains a broad combinatorial component as well as a visible accumulation near the expected B-meson mass region. This broad structure is typical of a preselection-level hadron-collider sample, where many random track combinations can satisfy a loose candidate definition.

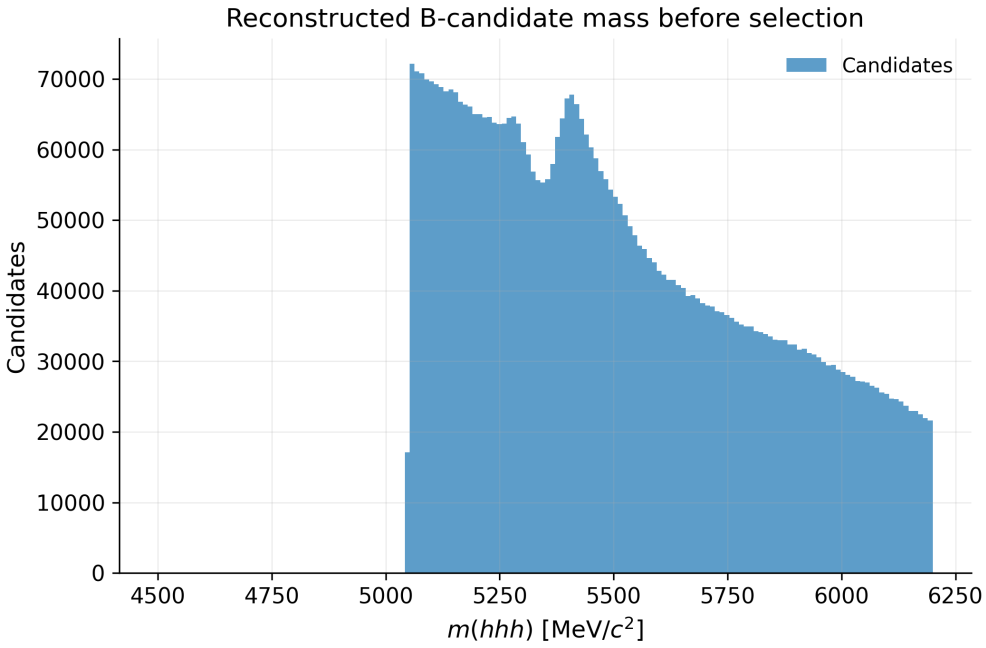


Figure 1: Reconstructed B-candidate invariant-mass distribution before baseline selection.

After applying quality, displacement, and kaon-PID selections, the mass spectrum becomes visibly cleaner, as shown in Fig. 2. The selection enhances the relative contribution of candidates compatible with a displaced B decay and suppresses a large fraction of combinatorial background.

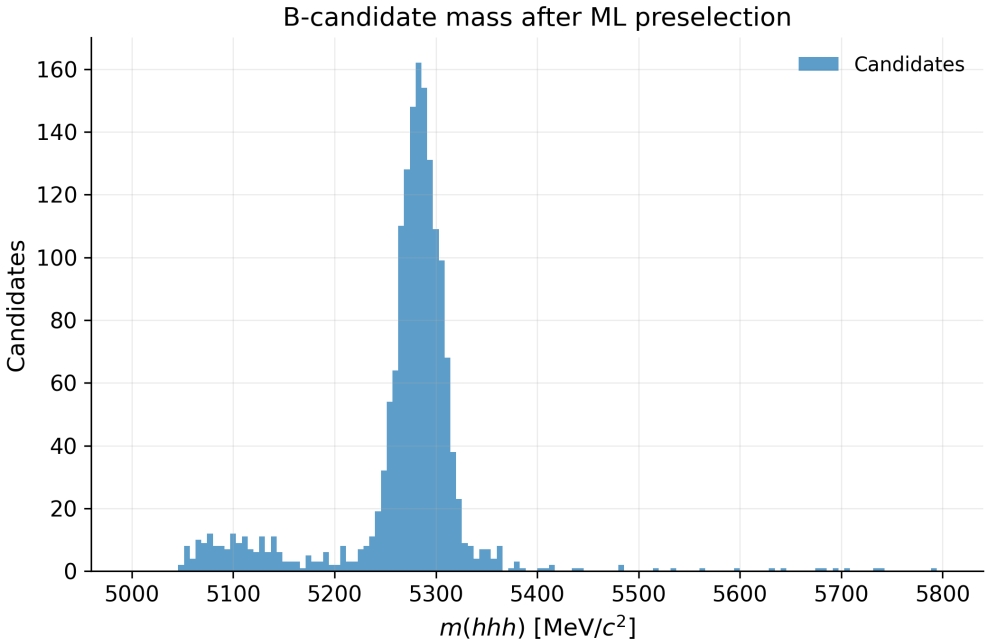


Figure 2: Reconstructed B-candidate invariant-mass distribution after baseline physics selections.

4.3 Signal-plus-background fit

A nominal signal extraction is performed with an extended signal-plus-background mass model,

$$f(m) = N_s S(m) + N_b B(m). \tag{7}$$

Here $S(m)$ is a common-mean double-Gaussian signal model for the reconstructed B-candidate peak, $B(m)$ is a finite-range exponential background model for the combinatorial component, and N_s and N_b are the fitted signal and background yields in the fitted mass interval. This model is used as the nominal mass-fit configuration later for the charge-separated B^+ and B^- fits. Alternative background descriptions are studied as systematic variations in the raw-asymmetry analysis, so the fit shown here should be interpreted as a nominal modelling choice rather than as a final physics measurement.

The fitted distribution is shown in Fig. 3. The visible peak near the B-meson mass demonstrates that the four-vector reconstruction and event selection are behaving sensibly, while the fitted background component provides a compact description of the residual combinatorial contribution.

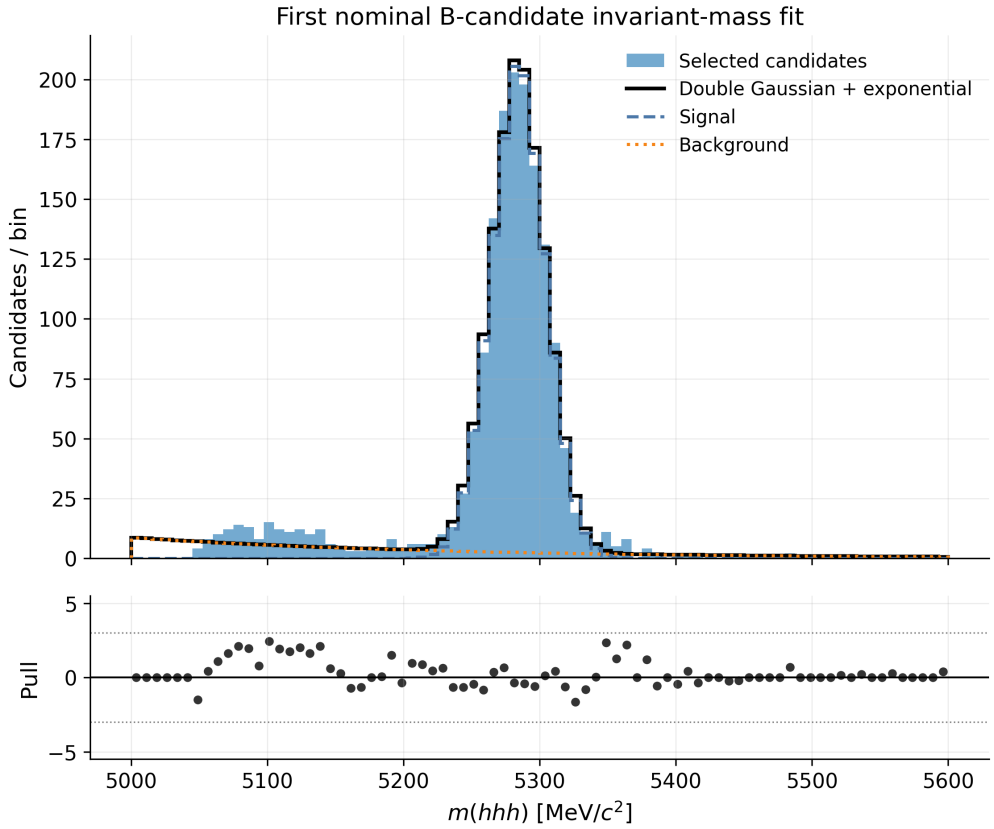


Figure 3: Selected B-candidate mass spectrum fitted with a double-Gaussian signal component and finite-range exponential background component.

5 Event Selection

Event selection is a central part of any hadron-collider analysis. The raw candidate sample contains true signal decays, partially reconstructed backgrounds, misidentified backgrounds, and random combinations of charged tracks. The goal of the baseline selection is to retain a useful signal-rich region while keeping the cuts simple enough to interpret.

The baseline selection used in this project is

$$\chi^2_{\text{vertex}}(B) < 5, \tag{8}$$

$$d_{\text{flight}}(B) > 20, \tag{9}$$

$$P_K(H_1), P_K(H_2), P_K(H_3) > 0.5. \tag{10}$$

Here $\chi^2_{\text{vertex}}(B)$ denotes the B-candidate vertex-quality variable, $d_{\text{flight}}(B)$ denotes the reconstructed flight-distance variable, and $P_K(H_i)$ denotes the kaon PID response for daughter track

H_i.

The vertex-quality cut suppresses candidates whose three tracks do not form a well-reconstructed secondary vertex. This is physically motivated because the daughters of a true B decay should originate from a common decay point. A poor vertex fit is more likely to arise from accidental combinations of tracks produced in different processes or from poorly reconstructed tracks.

The flight-distance selection exploits the finite lifetime of B mesons. Because B hadrons travel a measurable distance before decaying, their decay vertices are displaced from the primary proton–proton interaction point. Requiring a sufficiently large flight distance therefore suppresses prompt background combinations that originate near the primary vertex.

The PID selection implements the kaon mass hypothesis used in the invariant-mass reconstruction. Requiring all three tracks to have kaon-compatible PID values improves the consistency of the reconstructed final state. In a more complete analysis, one would optimize PID requirements with calibration samples and evaluate the corresponding efficiency and misidentification rates. Here the selection is chosen as a transparent demonstration of how detector-level PID information enters the analysis.

The selected sample is used consistently across the mass reconstruction, signal/background classification, CP-asymmetry diagnostic, and Dalitz-plot studies. This consistency is important: changing the selection between notebooks would make comparisons ambiguous and could introduce artificial differences in the resulting distributions.

6 Signal and Background Classification

6.1 Sideband-labelled samples

The signal/background study uses mass windows to create approximate labels. Candidates in a narrow window around the B mass are labelled as signal-like, while candidates in lower and upper mass sidebands are labelled as background-like. The sideband method is common in HEP because sideband regions provide a data-driven representation of combinatorial background.

The label definition used in the notebook is

$$\text{signal}: \quad 5270 < m_B < 5290 \text{ MeV}/c^2, \tag{11}$$

$$\text{background}: \quad 5200 < m_B < 5250 \text{ MeV}/c^2 \quad \text{or} \quad 5310 < m_B < 5350 \text{ MeV}/c^2. \tag{12}$$

These labels are proxy labels. The signal window can still contain background, and the sidebands may not perfectly represent the background under the signal peak. Nevertheless, the construction is useful for demonstrating a realistic machine-learning workflow without relying on simulation truth information.

6.2 Random Forest classifier

A Random Forest classifier is trained using reconstruction-quality and PID variables. The B-candidate mass is deliberately excluded from the feature list, because the mass defines the labels. Including it would make the classifier learn the sideband construction rather than physically meaningful differences in candidate quality or PID response.

The classifier uses the following feature set:

$$\left\{\chi^2_{\text{vertex}}(B),\, d_{\text{flight}}(B),\, P_K(H_1), P_K(H_2), P_K(H_3),\, P_\pi(H_1), P_\pi(H_2), P_\pi(H_3)\right\}. \tag{13}$$

The trained model is evaluated on a held-out test sample. The receiver operating characteristic (ROC) curve in Fig. 4 shows the trade-off between signal efficiency and background misclassification rate. The diagonal line corresponds to a classifier with no discriminating power.

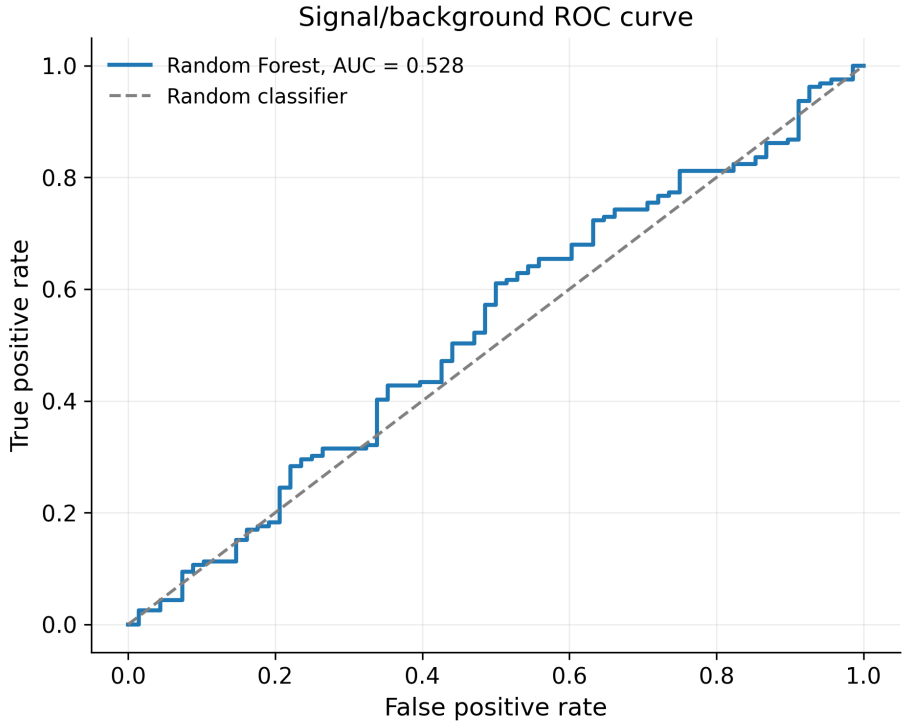


Figure 4: ROC curve for the sideband-labelled Random Forest signal/background classifier.

Figure 5 shows the feature-importance ranking from the Random Forest model. Such rankings are useful diagnostics, but they should not be overinterpreted as direct physics observables. They depend on feature correlations, sample composition, and the chosen model architecture.

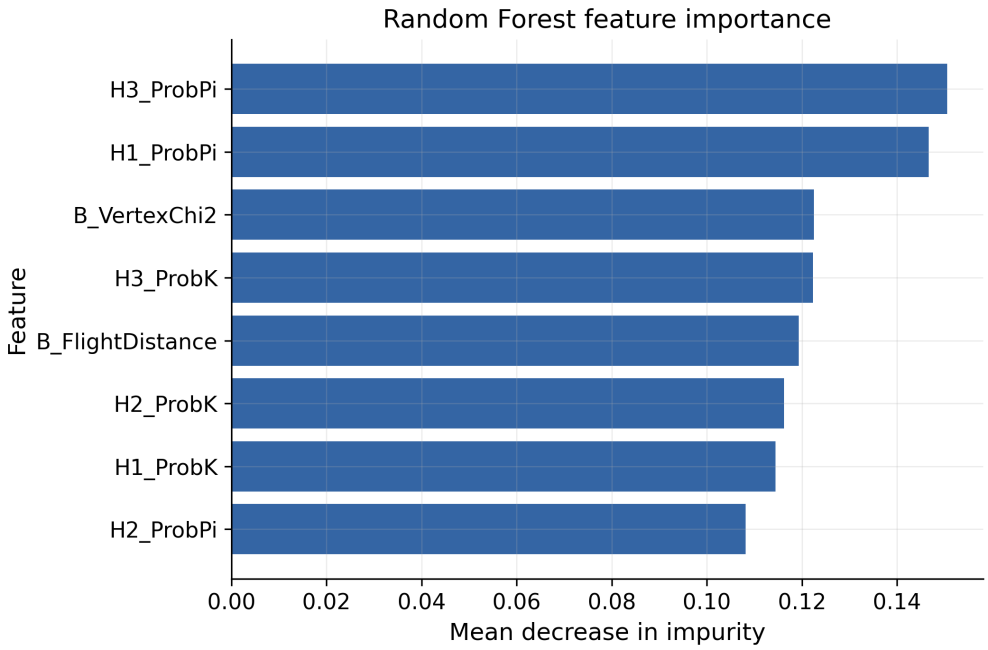


Figure 5: Random Forest feature importance for the signal/background classification task.

This part of the project demonstrates the integration of standard HEP sideband logic with modern machine-learning tools. Future improvements could include gradient-boosted decision trees, cross-validation, systematic studies of the sideband definition, and comparisons with simulation.

7 CP Asymmetry Study

The selected candidates can be separated into charge-conjugate samples using the total charge of the three daughter tracks,

$$Q_{\mathrm{tot}} = Q(H_1) + Q(H_2) + Q(H_3). \tag{14}$$

Candidates with $Q_{\mathrm{tot}} = +1$ are classified as B^+ candidates, while candidates with $Q_{\mathrm{tot}} = -1$ are classified as B^- candidates. This assignment uses only reconstructed track-charge information and is therefore a natural first step toward a charge-asymmetry study in a three-charged-hadron final state.

7.1 Charge-separated invariant-mass fits

Rather than counting candidates in a fixed signal window, the updated asymmetry extraction fits the B^+ and B^- invariant-mass spectra separately. Each charge sample is fitted in the range $5000 < m(hhh) < 5600$ MeV/c^2 with a common-mean double-Gaussian signal model and a finite-range exponential background model. The double-Gaussian signal shape provides a narrow core plus a broader resolution component, which is a more realistic detector-level description than a single Gaussian.

The fitted spectra are shown in Fig. 6 and Fig. 7. The fits converge for both charge categories and give acceptable goodness-of-fit values, with $\chi^2/\mathrm{ndof} = 1.42$ for the B^+ sample and $\chi^2/\mathrm{ndof} = 0.79$ for the B^- sample. The fitted signal yields are

$$N(B^+) = 694.4 \pm 27.2, \tag{15}$$

$$N(B^-) = 640.0 \pm 25.8, \tag{16}$$

where the uncertainties are the statistical yield uncertainties returned by the binned mass fits.

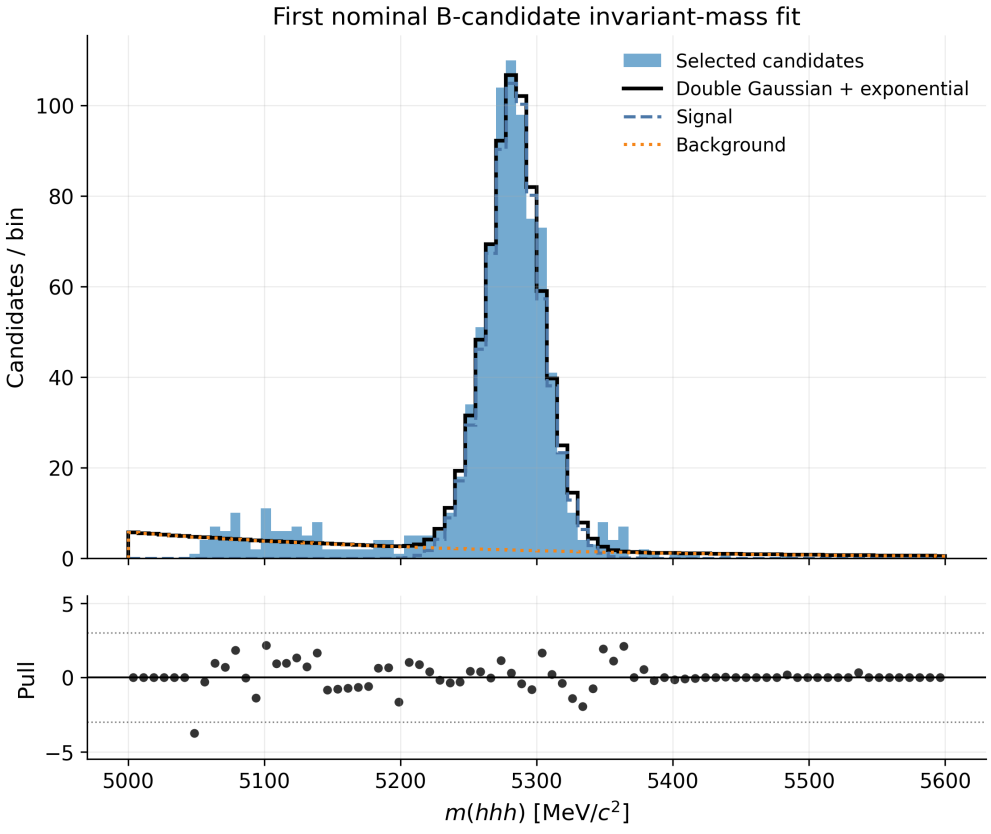


Figure 6: Charge-separated invariant-mass fit for the B^+ candidate sample.

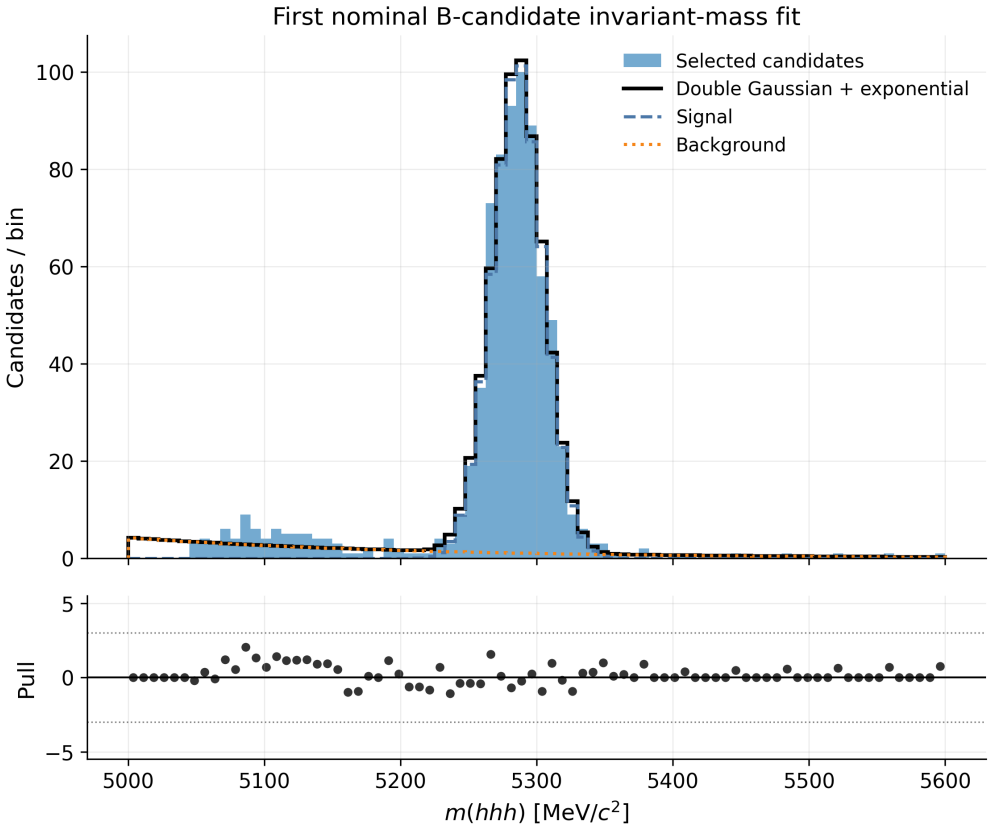


Figure 7: Charge-separated invariant-mass fit for the B^- candidate sample.

7.2 Raw asymmetry extraction

The fitted signal yields are used to construct the raw asymmetry

$$A_{\text{raw}} = \frac{N(B^-) - N(B^+)}{N(B^-) + N(B^+)}. \tag{17}$$

Propagating the independent fitted-yield uncertainties gives

$$A_{\text{raw}} = -0.0408 \pm 0.0281 \text{ (stat.)}. \tag{18}$$

After including the systematic variations described below, the result is reported as

$$A_{\text{raw}} = -0.0408 \pm 0.0281 \text{ (stat.)} \pm 0.0433 \text{ (syst.)}. \tag{19}$$

This is a raw asymmetry, not a physical CP-asymmetry measurement. It still contains detector-induced charge asymmetries, possible B^+/B^- production asymmetries, trigger and reconstruction effects, residual background-model dependence, and selection-induced charge biases. A complete measurement would require detector calibration, magnet-polarity combination, background control, efficiency corrections, and a more complete systematic uncertainty programme. The value in Eq. 19 should therefore be interpreted as a diagnostic of the analysis workflow rather than as evidence for CP violation.

7.3 Toy-MC validation

The statistical machinery was validated with pseudo-experiments generated from the nominal charge-separated fitted model. In each toy experiment, B^+ and B^- mass spectra were generated with the injected fitted yields and then refitted with the same double-Gaussian plus exponential

model used in the data fit. The toy study checks whether the fitting and uncertainty propagation recover the injected raw asymmetry when the fit model is correct.

A 100-toy validation with fixed random seed produced 99 toys suitable for pull studies and no fit failures. For the valid toys, the mean raw-asymmetry pull was -0.049 and the pull width was 1.052, compatible with the expected standard normal behaviour for this sample size. The fitted bias was -0.0025, well below the typical fitted statistical uncertainty of about 0.029. These results indicate that the nominal statistical extraction is well behaved on pseudo-data generated from the fitted model. This validation is not a detector-level systematic study.

7.4 Systematic uncertainties

Two first systematic categories are evaluated for the raw asymmetry: the choice of mass-fit model and the choice of invariant-mass fit range. In both cases, alternative fits are accepted only if the B^+ and B^- fits converge, the fitted yields and yield uncertainties are positive, and the χ^2/ndof values remain below 2.5 for both charge-separated mass spectra.

The fit-model study compares the nominal double-Gaussian plus exponential model with an accepted double-Gaussian plus linear-background alternative. A single-Gaussian plus exponential model is also retained as a stress test, but it is not used to assign the systematic uncertainty because it does not provide an acceptable description of the B^- spectrum. The accepted linear-background variation shifts the raw asymmetry by 0.0255, which is taken as the fit-model systematic uncertainty.

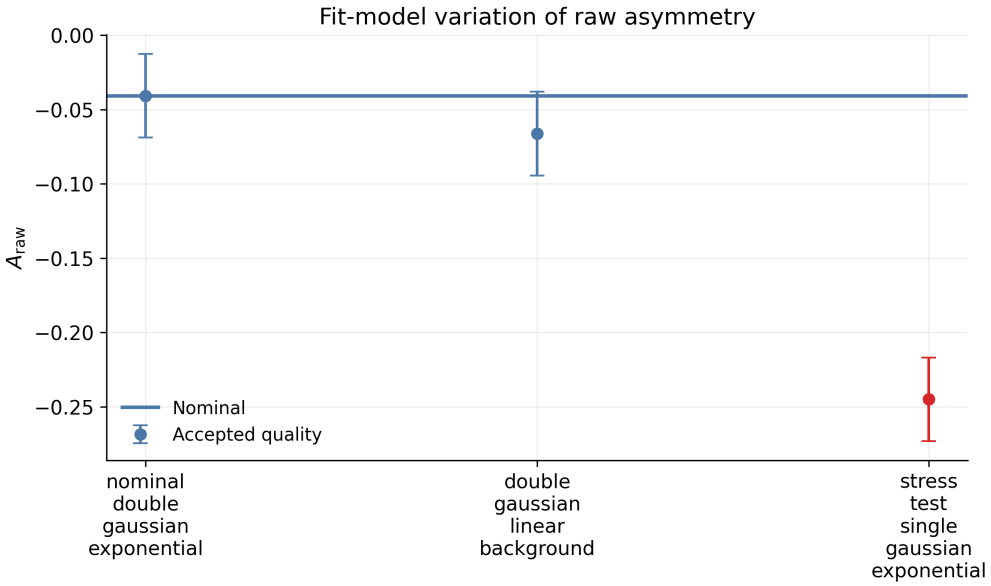


Figure 8: Raw asymmetry obtained with different charge-separated mass-fit models. Rejected stress-test variations are shown for comparison but are not used in the systematic assignment.

The fit-range study repeats the nominal double-Gaussian plus exponential fits for narrower and wider mass intervals. These variations test the sensitivity to the amount of sideband information included in the background constraint. All tested fit-range variations pass the quality requirements. The largest accepted shift is 0.0350, obtained from the wider high-mass range $5000 < m(hhh) < 5700$ MeV/c^2, and is assigned as the fit-range systematic uncertainty.

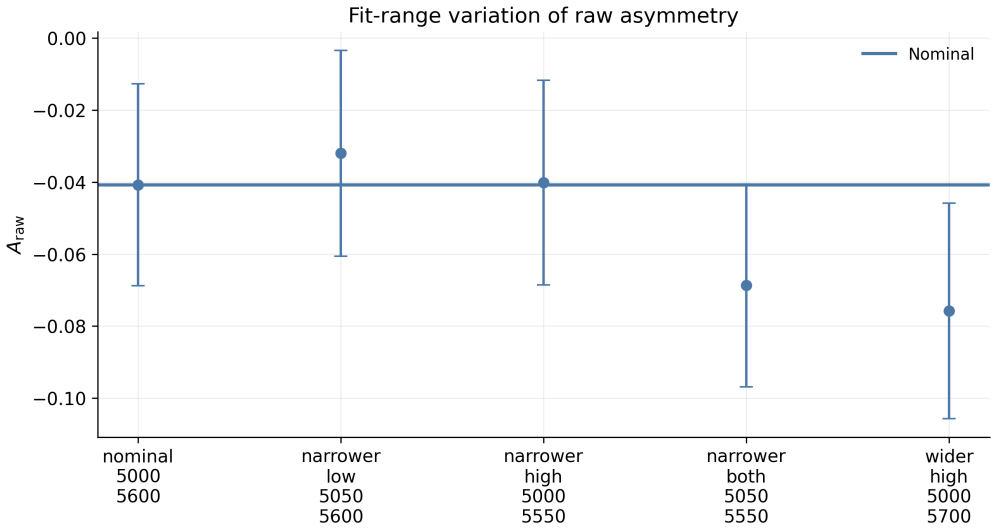


Figure 9: Raw asymmetry obtained for alternative invariant-mass fit ranges using the nominal signal and background model.

Table 2 summarizes the uncertainty components. The total systematic uncertainty is obtained by adding the fit-model and fit-range components in quadrature. The total uncertainty, including statistics, is 0.0516.

Table 2: Uncertainty components for the fitted raw asymmetry.

Source	Uncertainty on A_{raw}
Statistical	0.0281
Fit-model systematic	0.0255
Fit-range systematic	0.0350
Total systematic	0.0433

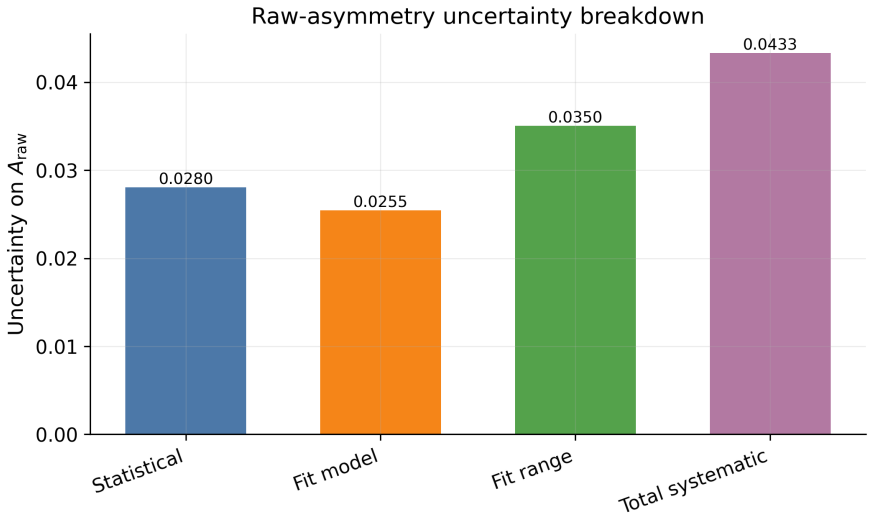


Figure 10: Breakdown of the statistical uncertainty, individual systematic components, and total systematic uncertainty on the raw asymmetry.

The dominant systematic contribution in this first study comes from the fit range, followed by the background-model variation. These studies probe only a limited set of analysis choices. Addi-

tional signal-shape variations, broader background-shape alternatives, magnet-polarity comparisons, and detector-efficiency corrections would be required before interpreting the raw asymmetry as a corrected CP observable.

8 Dalitz Plot Analysis

8.1 Pairwise invariant masses

Three-body decays contain more kinematic information than a single reconstructed parent mass. The internal phase-space structure can be studied using pairwise invariant masses. For three daughter tracks labelled K_1, K_2, and K_3, the Dalitz variables used in this project are

$$m^2(K_1K_2), \qquad m^2(K_2K_3), \qquad m^2(K_1K_3). \tag{20}$$

Only two of these variables are independent for a fixed parent mass, but plotting different pairs is useful for visual inspection and for checking symmetries or structures involving different track orderings.

The pair mass is computed by summing the two daughter four-vectors:

$$m_{ij}^2 = (E_i + E_j)^2 - |\vec{p}_i + \vec{p}_j|^2 \,. \tag{21}$$

As in the B-mass reconstruction, each daughter is assigned the kaon mass hypothesis. The resulting squared masses are reported in GeV^2/c^4 to keep the numerical axis scales readable.

8.2 Dalitz distributions and resonance structure

A Dalitz plot displays the density of selected events in a two-dimensional plane of pairwise invariant-mass-squared variables. Resonant two-body substructures can appear as bands, while interference and phase-space effects can produce non-uniform densities. In a complete amplitude analysis, these structures would be modelled with coherent sums of resonant and non-resonant amplitudes. The present analysis restricts itself to visualising the distributions.

Figure 11 shows the distribution in the $m^2(K_1K_2)$ versus $m^2(K_2K_3)$ plane. The use of a two-dimensional histogram rather than a raw scatter plot makes dense regions easier to read.

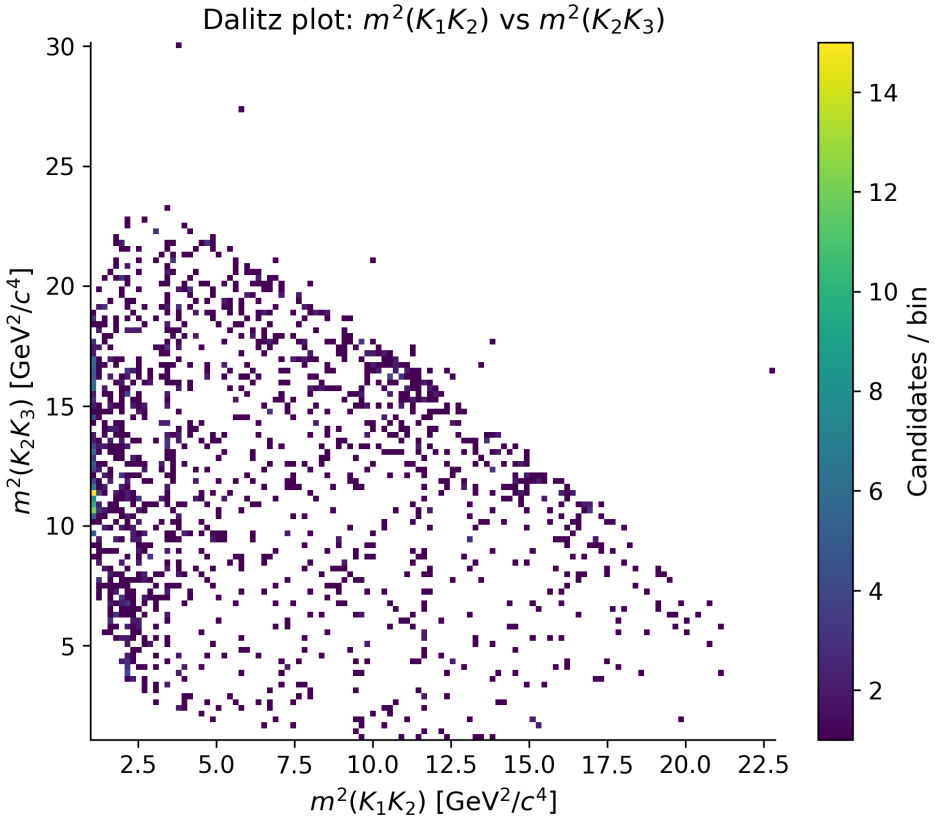


Figure 11: Dalitz density map in the $m^2(K_1K_2)$ versus $m^2(K_2K_3)$ plane.

Figures 12 and 13 show the complementary pairings. Together, these projections provide a first view of the three-body phase-space population after the baseline selection.

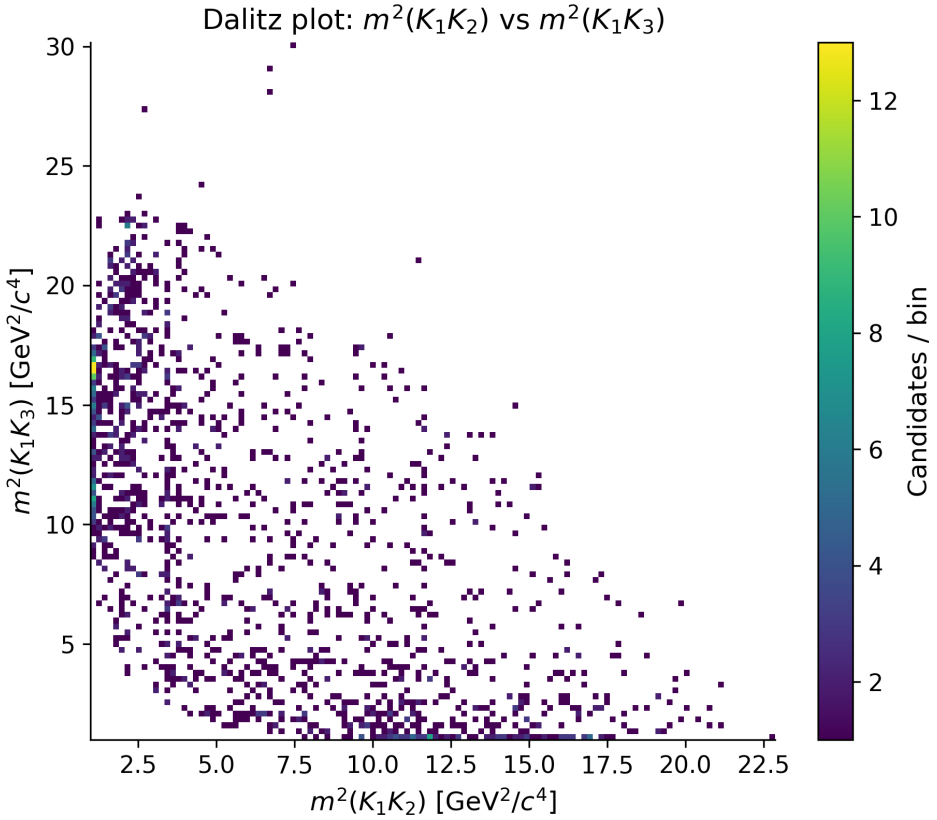


Figure 12: Dalitz density map in the $m^2(K_1K_2)$ versus $m^2(K_1K_3)$ plane.

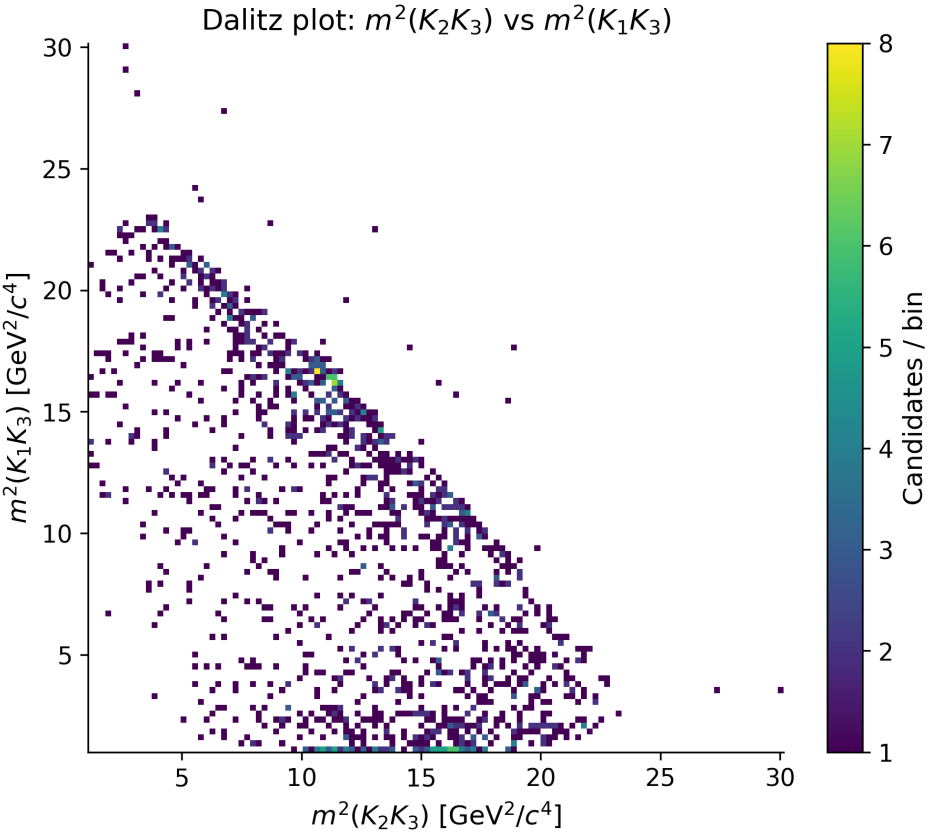


Figure 13: Dalitz density map in the $m^2(K_2K_3)$ versus $m^2(K_1K_3)$ plane.

Dalitz plots are especially relevant for CP-violation studies in multibody decays. Even if the integrated raw asymmetry is small, local asymmetries in restricted phase-space regions can be larger. This motivates a first bin-by-bin diagnostic of charge-dependent structures across the Dalitz plane.

8.3 Local raw-asymmetry diagnostic

The local diagnostic uses the same selected candidates as the integrated raw-asymmetry study. Candidates are split into B^+ and B^- samples using the total reconstructed daughter charge,

$$q_{\mathrm{tot}} = q_1 + q_2 + q_3. \tag{22}$$

The $m^2(K_1K_2)$ versus $m^2(K_2K_3)$ plane is then divided into regular two-dimensional bins. In each bin (i, j), the raw asymmetry is computed directly from candidate counts,

$$A_{\mathrm{raw}}^{ij} = \frac{N_-^{ij} - N_+^{ij}}{N_-^{ij} + N_+^{ij}}, \tag{23}$$

where N_+^{ij} and N_-^{ij} are the numbers of B^+ and B^- candidates in that Dalitz bin. The statistical uncertainty is estimated from binomial counting propagation, and a local significance is formed as $A_{\mathrm{raw}}^{ij}/\sigma(A_{\mathrm{raw}}^{ij})$.

The default study uses a 10×10 binning and masks bins with fewer than 20 selected candidates in the asymmetry and significance maps. The bin contents are still stored in the numerical output, but low-statistics regions are not interpreted visually. With this threshold, 26 out of 100 bins pass the requirement. The direct counting asymmetry over the selected candidates entering the local Dalitz study is

$$A_{\mathrm{raw}} = -0.0748 \pm 0.0246. \tag{24}$$

This number differs from the fit-based integrated asymmetry quoted in the CP-asymmetry section because it is a direct candidate-counting diagnostic. It does not subtract fitted background and does not correct for detector efficiency, phase-space acceptance, bin migration, production asymmetry, or detector-induced charge asymmetries.

Figure 14 shows the selected-candidate population used to define the local bins. The occupancy is highly non-uniform, reflecting both phase-space structure and the effect of the selection. This count map is essential context for interpreting the local asymmetry and significance maps.

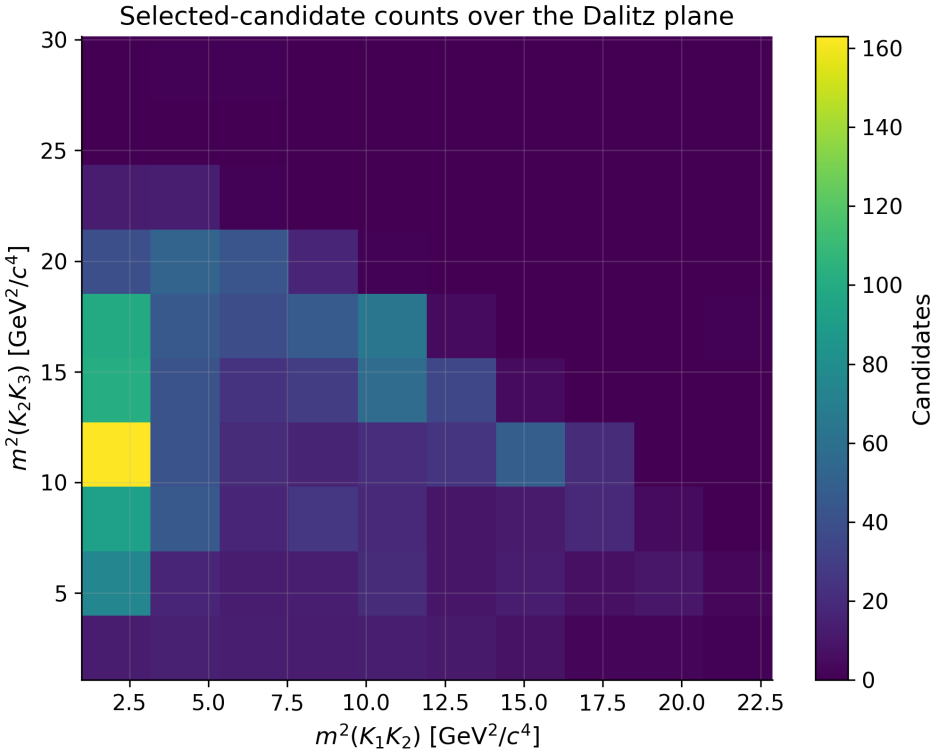


Figure 14: Selected-candidate counts in the $m^2(K_1K_2)$ versus $m^2(K_2K_3)$ Dalitz plane.

Figure 15 shows the local raw-asymmetry map after the minimum-count masking. The map is useful for visualising possible charge-dependent structures, but no physics claim is made from this diagnostic alone.

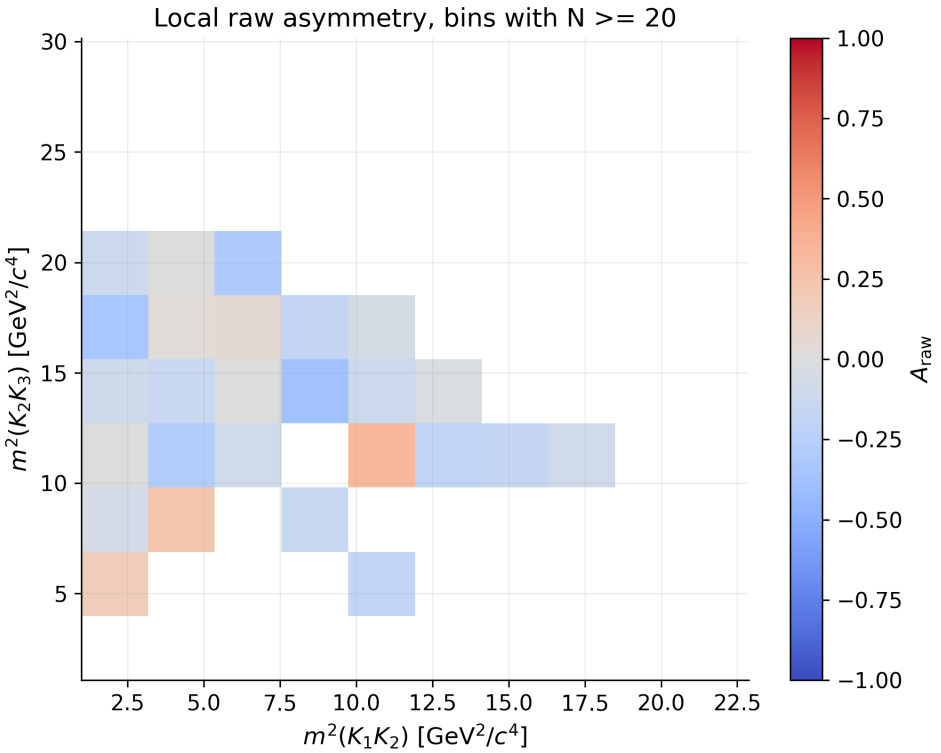


Figure 15: Local raw-asymmetry map in the $m^2(K_1K_2)$ versus $m^2(K_2K_3)$ Dalitz plane. Bins with fewer than 20 selected candidates are masked.

Figure 16 shows the corresponding local counting significance. The largest local deviations should be interpreted with caution because many bins are inspected and no look-elsewhere correction, background subtraction, or efficiency correction is applied.

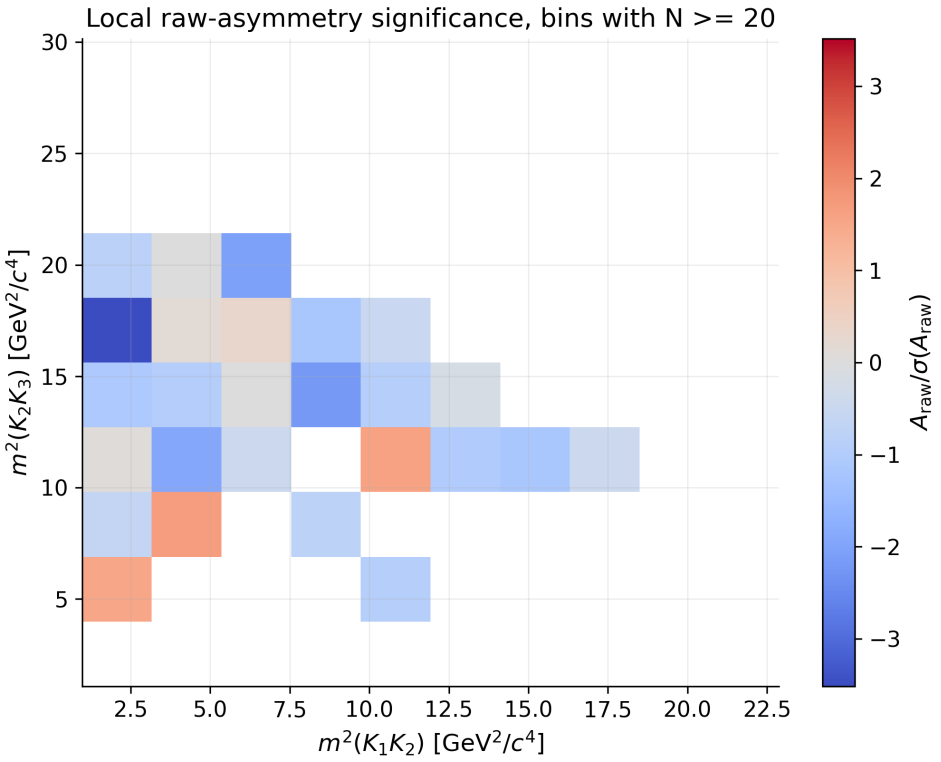


Figure 16: Local raw-asymmetry significance in the $m^2(K_1K_2)$ versus $m^2(K_2K_3)$ Dalitz plane. Bins with fewer than 20 selected candidates are masked.

A proper local CP-asymmetry study would require signal-yield extraction or background subtraction in each Dalitz region, efficiency and acceptance corrections, treatment of migration between bins, and detector- and production-asymmetry corrections. The maps shown here should therefore be read as a conservative exploratory diagnostic rather than as evidence for a physical CP asymmetry.

9 Conclusion

This project demonstrates a complete and modern introductory HEP analysis workflow using LHCb CERN Open Data. Starting from ROOT ntuples, the analysis reads selected branches with `uproot`, processes candidates with `pandas` and `numpy`, reconstructs invariant masses, applies physics-motivated selections, generates publication-style plots, trains a sideband-labelled Random Forest classifier, performs charge-separated invariant-mass fits, computes a raw charge asymmetry, and visualises three-body phase space with Dalitz plots.

The invariant-mass reconstruction shows a clear B-candidate peak after the baseline selection. The selected mass spectrum is described with a double-Gaussian signal model and an exponential background model, providing a realistic detector-level signal extraction framework. The signal/background classification study demonstrates how sideband labels can be combined with standard machine-learning tools, while the ROC curve and feature-importance plot provide useful performance diagnostics. The CP-asymmetry study separates B^+ and B^- candidates, performs independent invariant-mass fits, and extracts a raw integrated asymmetry together with statistical and systematic uncertainty estimates. Toy-MC pseudo-experiments are used to validate the statistical behaviour of the fitting procedure and uncertainty propagation. The Dalitz analysis extends the project toward multibody-decay phase-space studies by reconstructing pairwise invariant-mass-squared variables.

The analysis remains intentionally simplified and does not constitute a precision flavour-physics

measurement. Several improvements would still be required for a full experimental analysis:

- train stronger classifiers such as XGBoost or gradient-boosted decision trees, with cross-validation and robust overtraining checks;
- include detector effects, PID calibration, trigger efficiencies, and reconstruction asymmetries;
- use simulation and control samples to validate selection efficiencies and background shapes;
- combine MagnetUp and MagnetDown samples to reduce charge-dependent detector biases;
- implement additional accepted fit-model variations, such as Crystal Ball signal models or alternative background parameterisations;
- perform local CP-asymmetry studies across Dalitz-plot regions.

Despite these limitations, the project captures several essential components of a realistic experimental-particle-physics analysis: transparent event reconstruction, documented selection choices, modular code, reproducible figures, signal extraction through invariant-mass fitting, uncertainty propagation, toy-MC validation, and explicit treatment of systematic effects. It therefore provides a strong foundation for more advanced studies in heavy-flavour physics and for future work in an experimental PhD environment.

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